

# The Thermodynamic Mandate and Dissipative Structuring: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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## Abstract

This manuscript frames abiogenesis as a nonequilibrium systems problem rather than a single lucky molecular accident. Constraint-Driven Flux Dynamics (CDFD) is used as a coarse-grained language for asking when environmental energy flux, geometric constraint, active surface formation, and memory-like persistence can reinforce one another. The central object is the adaptive operating ratio  $\Psi_s = (\Phi/C) \cdot S \cdot M_s$ , where  $\Phi$  denotes available driving flux,  $C$  denotes the effective resistance or constraint field,  $S$  denotes responsive organization, and  $M_s$  denotes retained structural memory. This is the first paper in a twelve-paper series; it does not prove that life is inevitable in every driven environment, but establishes the dependency structure that later papers test in geochemical, polymerization, compartment, replication, parasitic, photochemical, and synthesis settings.

**Keywords:** origins of life; nonequilibrium thermodynamics; dissipative structures; constraint-driven dynamics; computational modelling.

## Preprint note

This is a computational research article. The companion script produces numerical diagnostics reported in the Computational Results section. It is not a narrative review and does not claim historical proof of a unique abiogenesis pathway.

## Symbol Conventions

This is the reference paper for the full CDFD origins-of-life symbol convention used throughout Part II. Every other paper in the series points back here rather than repeating the full tables.  
**CDFD origins-of-life symbol convention.**

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| Symbol                  | Meaning in Part II                                | Origins-of-life reading                                                                                                      |
|-------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $\Phi$                  | Available driving flux                            | Redox, thermal, photochemical, proton/ion, or chemical potential drive supplied by the environment                           |
| $C$                     | Effective constraint or resistance                | Mineral geometry, pore confinement, diffusion barrier, permeability, activation barrier, or maintenance burden               |
| $S$                     | Responsive organization                           | Surface, boundary, catalytic network, coacervate, or protocellular structure that routes flux into retained function         |
| $M_s$                   | Retained structural memory                        | Composition, sequence, topology, compartment state, mineral imprint, or boundary history that persists across renewal cycles |
| $\Psi_s$                | Local adaptive ratio $(\Phi/C) \cdot S \cdot M_s$ | Bookkeeping gauge for whether drive, constraint, responsiveness, and retention are co-present; not a universal constant      |
| $\Lambda_{\text{life}}$ | Integrated Life Number where used                 | Capstone surplus ratio for decay, near-critical proto-biology, and sustained self-maintenance (defined in Paper 12)          |

### Greek-symbol convention.

| Symbol                     | Local role                                                               | Interpretation rule                                                                                     |
|----------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $\alpha$                   | Accumulation or reinforcement coefficient                                | Defines how fast a constraint, pathway, or retained structure grows under load                          |
| $\beta$                    | Relaxation, decay, or washout coefficient                                | Defines recovery, loss, clearance, hydrolysis, or return toward baseline                                |
| $\gamma$                   | Coupling, diffusion, or redistribution coefficient                       | Defines spread across pores, surfaces, compartments, networks, or chemical phases                       |
| $\delta, \epsilon$         | Perturbation, tolerance, or small offset scales                          | Used only as local thresholds, numerical guards, or experimental tolerances                             |
| $\eta, \lambda, \mu, \nu$  | Efficiency, rate, baseline, or frequency terms                           | Meaning is set by the equation and subscript where they occur                                           |
| $\rho, \sigma, \tau, \phi$ | Density/correlation, conductance/noise, time-scale, or phase/field terms | Never universal constants unless a paper explicitly defines them that way                               |
| $\Omega, \Lambda$          | Domain/state space; Life Number where used                               | $\Omega$ names a modeled state space; $\Lambda$ is reserved for life-number notation only where defined |

**Working-convention status.**  $\Psi_s = \Phi/C$  and the four-term product  $\Psi_s = (\Phi/C) \cdot S \cdot M_s$  are adopted here as a *working notational convention* for Part II bookkeeping, not as a derived physical law from Part I. The biology series remains evaluable on its own chemical and geological evidence regardless of how Part I revises the status of that ratio.

**Note on the  $\Psi_s$  closure across this series.** Several papers in this sequence (Papers 2, 3, and 4) restrict discussion to the ratio  $\Phi/C$  alone before  $S$  and  $M_s$  are formally developed in Papers 6–7. Where this occurs, it should be read as a deliberate structural restriction— $S$  and

$M_s$  are held implicitly at their neutral values until the mechanisms that give them content are introduced—not as an inconsistency in the master equation. Papers 1, 5, 8, and 12 use the full four-term product.

## 1 Introduction

Living systems maintain local order by coupling themselves to external gradients. The origin-of-life problem is therefore not only the problem of making particular molecules; it is also the problem of sustaining useful work against dilution, hydrolysis, side reactions, and spatial loss. In CDFD language, a prebiotic system becomes interesting when a flux source and a constraint architecture create a persistent opportunity for organization.

This manuscript supplies the series-level vocabulary — CDFL, the CDFD flow-constraint-memory language used by the public runtime and by the manuscript series. It deliberately keeps the claim at the level of mechanism class: driven systems can favor structures that increase usable dissipation, but only if those structures also retain enough organization to persist.

**Relation to prior dissipation-based accounts of abiogenesis.** This framing sits inside an existing thermodynamic-dissipation lineage in origins-of-life research and should be read as continuous with it, not as an independent discovery. Prigogine’s dissipative-structures program established that systems maintained far from equilibrium can spontaneously organize to increase entropy production [12] — the source of the phrase used in this paper’s title. England’s statistical-physics account of self-replication [3] makes a closely related claim in modern notation: driven, dissipative systems can be statistically favored to adopt structures that absorb and dissipate more of the available driving work, which is essentially the CDFD  $\Psi_s$  claim restated with a different formalism. Michaelian’s thermodynamic dissipation theory for the origin of life [7, 8] argues for a directly competing account centered on UV-driven dissipation as the primary selective force at the origin of life. CDFD’s contribution relative to these three is a shared bookkeeping notation across geochemical, polymerization, and protocellular stages, rather than a new physical mechanism — the underlying thermodynamic intuition is not new, and later work in this series should be read as an application of that established framework rather than a departure from it.

The same flux-constraint grammar is also applied elsewhere in the CDFD program (fundamental physics and, prospectively, medicine). Origins-of-life claims in Part II are expected to be judged on their own chemical and geological evidence regardless of what happens to the formalism in those other domains.

## 2 Scientific Lineage and Complementarity

This paper stands on a long origin-of-life tradition rather than replacing it. Oparin’s coacervate program, Wächtershäuser’s surface-metabolism proposal, Russell and Martin’s alkaline hydrothermal-vent work, Eigen and Orgel’s error-threshold logic, Kauffman’s autocatalytic-set theory, Szostak’s protocell program, Lane’s bioenergetic arguments, Powner’s nucleotide chemistry, Patel’s cyanosulfidic network chemistry, and Damer’s cycling-environment hypothesis all

identify real parts of the puzzle [15, 13, 6, 2, 9, 4, 14, 5, 11, 10, 1]. CDFD’s role is to provide a single dependency grammar across those traditions: what is the input, what constrains it, what routes it, what is retained, and what would falsify the claimed coupling? This framing is meant to complement those schools, not erase their disagreements.

### 3 Notation independence

Shared CDFD/CDFL notation here is a disciplined bookkeeping layer only:  $\Phi$  names drive,  $C$  names resistance or constraint,  $S$  names responsive organization, and  $M_s$  names retained structure. As noted under Symbol Conventions, the  $\Psi_s$  product is a working convention for this biology series rather than a load-bearing derivation from any other CDFD installment. Evaluation of Part II stands on chemical and geological evidence alone.

### 4 Flux, Constraint, Surface, and Memory

The shared phenomenological balance used throughout this Part II sequence is

$$\frac{\partial \Phi}{\partial t} = \nabla \cdot \left( \frac{1}{C} \nabla \Phi \right) + S - D, \quad (1)$$

where  $D$  collects loss, degradation, washout, and uncoupled dissipation. The terms are interpreted as follows:

- **Flux** ( $\Phi$ ): chemical, redox, photochemical, thermal, or electrochemical drive available to the system.
- **Constraint** ( $C$ ): transport resistance, activation barrier, geometric confinement, permeability, or maintenance burden.
- **Responsiveness** ( $S$ ): the ability of local structure to route flux into productive pathways rather than noise.
- **Systemic memory** ( $M_s$ ): persistence of structure, sequence, composition, topology, or boundary state across time.

All symbols are local phenomenological variables unless explicitly tied to a measurement. They should not be read as universal constants. Later papers redefine specialized symbols only after stating their physical meaning, because the same letter can otherwise mean different things in reaction kinetics, transport theory, and bioenergetics.

The adaptive ratio

$$\Psi_s = \left( \frac{\Phi}{C} \right) S M_s \quad (2)$$

is not introduced as a universal constant. It is a bookkeeping device for tracking whether the components needed for persistence are co-present. A high flux without retention burns out. A strong constraint without flux freezes. A responsive surface without memory remains transient.

## 5 Necessary Conditions for the Part II Model

The Part II claim holds only if four necessary conditions are met together. They are stated explicitly because origin-of-life narratives often over-weight one ingredient: a favored molecule, a favored setting, or a favored energy source. CDFD treats those ingredients as incomplete unless they satisfy a coupled gate.

1. **Sustained disequilibrium.** The environment must supply repeatable drive, not a single pulse. Operationally,  $\Phi$  must remain positive over the relevant renewal interval.
2. **Finite constraint.** The system must be neither freely mixed nor sealed off. In symbols,  $0 < C < \infty$ : too little constraint gives washout, while too much constraint prevents exchange and renewal.
3. **Responsive routing.** The local structure must alter product distributions or retention. A surface, pore, droplet, or polymer has CDFD significance only when it changes where flux goes, represented by  $S > 0$ .
4. **Retained memory.** Some product, pattern, boundary state, or catalytic bias must survive later destructive phases. The working test is  $M_s(t + nT) > M_s(t)$  over at least one repeated environmental sequence.

## 6 Computational Results: Dependency-Gate Diagnostic

The companion script `supplementary_ool_01.py` scans a two-dimensional grid of drive  $\Phi$  and constraint  $C$ , then computes Eq. (2) using bounded phenomenological functions for  $S$  and  $M_s$ . **This is an illustrative diagnostic, not a validated Results section:** it is a necessary-condition map, not a simulation of life, and the specific numbers it reports depend on the toy script’s chosen parameter ranges rather than any measured chemistry.

The scan reports 3,600 grid points, of which 425 fall in the near-critical window  $0.85 \leq \Psi_s \leq 1.15$ . That window’s boundaries are chosen for illustration and have not been shown to be robust to alternative reasonable choices (e.g.,  $\pm 10\%$  or  $\pm 20\%$  instead) — a sensitivity sweep confirming the qualitative three-regime picture across several window widths is identified here as required future work, not yet performed. The diagnostic’s purpose is to force the release to distinguish three regimes:

- underdriven or overconstrained systems, where chemistry cannot build persistent memory;
- near-critical systems, where later mechanisms may matter;
- overloaded systems, where excess drive outruns retention or control.

## 7 Dissipative Structuring

Driven systems can form structures that route energy more effectively than homogeneous diffusion. In prebiotic settings, the relevant question is whether such structures can become chem-

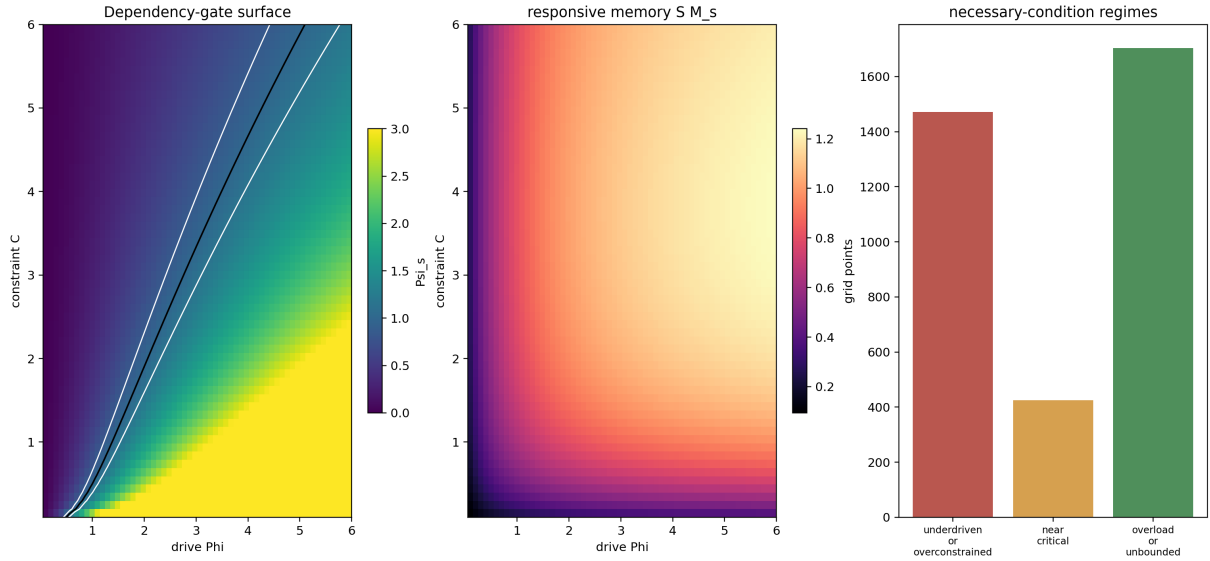


Figure 1: Dependency-gate diagnostic. The near-critical band marks where drive, finite constraint, responsive routing, and retained memory are co-present in the toy surface. It is an interpretive map for the twelve-paper chain, not evidence that abiogenesis has occurred.

ically consequential: mineral interfaces, pores, films, droplets, polymers, and reaction loops all change the effective relationship between  $\Phi$  and  $C$ .

For a bare chemical soup, one may have  $S \approx 0$  and  $M_s \rightarrow 0$ , so transient fluctuations do not accumulate. A successful origin pathway must raise both quantities without disconnecting from the external drive. The next concrete step is geological boundary conditions: the framework needs a specific geochemical substrate before it becomes a concrete origin-of-life argument, developed in Paper 2.

## 8 Claim Boundary

The strongest defensible claim of this paper is conditional:

If a prebiotic environment supplies sustained disequilibrium, spatial constraint, and chemistry capable of retaining useful structure, then CDFD predicts selection for configurations that improve coupled throughput and persistence.

This is not a proof that a particular vent, pool, mineral, or molecule was historically necessary. The later papers narrow those possibilities by adding Fe–S redox chemistry, distributed mineral transport, proton/ion coupling, cycling polymerization, autocatalysis, chemical alphabets, compartments, replication thresholds, parasitic failure modes, and photochemical expansion.

## 9 Series-Level Falsification Logic

The present analysis can fail in several ways. If prebiotic experiments show that product persistence is independent of constraint architecture, then  $C$  is a weak or decorative variable.

If surfaces change flux but not endpoint chemistry, then  $S$  is not a useful organizing parameter. If repeated cycles do not improve retained product distributions relative to matched steady controls, then  $M_s$  does not describe chemical memory in the relevant setting. Finally, if each paper in the series requires a different incompatible definition of  $\Psi_s$ , the CDFL grammar fails as a unifying language even if individual origin-of-life mechanisms remain interesting.

## 10 Conclusion

The thermodynamic opening move is a dependency graph, not a victory lap. Life-like organization requires flux, constraint, responsive routing, and retained memory to reinforce one another. A natural continuation examines the most immediate physical substrate: iron–sulfur redox and mineral constraint scaffolds (Paper 2).

## 11 Reproducibility Note

The companion script `supplementary_ool_01.py` writes `dependency_gate_surface.png` under `outputs/paper_01/`, plus a grid CSV, a necessary-conditions CSV, and summary JSON. The output is a necessary-condition visualization, not validation of abiogenesis.

## Related literature

This manuscript defines the Part II thermodynamic entry point on chemical and geological grounds. Shared CDFD/CDFL symbols are introduced locally as bookkeeping; Part I vacuum-physics papers are not load-bearing here. Field anchors: Prigogine and Nicolis [12]; England [3]; Michaelian [7, 8].

## Competing Interests

The author declares no competing interests.

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## Data and Code Availability

Companion scripts, generated figures, and numerical outputs for this manuscript are included in the public CDFD Part II release archive (<https://github.com/bampita-bico/CDFD-Part-II-Release>; Zenodo DOI 10.5281/zenodo.20264779). See the Reproducibility Note above for the paper-local script path.

## Author Contributions

Conceptualization, formal analysis, computational diagnostics, and manuscript preparation:  
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